Appendix B: Dashboard excerpt

templates/index.html

```
        <div class="chip"><div class="k">Tracked</div><div class="v" id="tracked">0</div></div>
        <div class="chip"><div class="k">Visible</div><div class="v" id="visible">0</div></div>
        <div class="chip"><div class="k">Feed</div><div class="v" id="status">Boot</div></div>
        <button id="mode-toggle" type="button">Simulation: ...</button>
        <div id="mode-badge" class="sim">? SIM</div>
      </div>
    </header>
    <section id="main">
      <aside id="listPane"><div id="listHdr">LIVE TRAFFIC</div><ul id="aircraftList"></ul></aside>
      <div id="mapWrap">
        <div id="map"></div>
        <div id="ctrl"><button id="zin">+</button><button id="zout">-</button><button id="center">o</button></div>
        <div id="legend">India sector radar</div>
        <div id="map-status">Offline basemap - live aircraft tracking active</div>
        <div id="live-banner"></div>
      </div>
    </section>
    <section id="bottom">
      <div id="bottom-resizer" title="Drag to resize details panel"><span>&#8597;</span></div>
      <div id="tabPane">
        <div id="tabs"><button data-tab="flight" class="active">Flight</button><button data-tab="aircraft">Aircraft</butto
| n><button data-tab="route">Route</button></div>
        <div id="flight" class="tab active"><div class="empty">Select an aircraft from map or list.</div></div>
        <div id="aircraft" class="tab"><div class="empty">No aircraft selected.</div></div>
        <div id="route" class="tab"><div class="empty">Route details appear on selection.</div></div>
      </div>
      <div id="graph"><canvas id="altitudeChart" height="170"></canvas></div>
    </section>
  </div>
  <script>
    const map=L.map("map",{zoomControl:false}).setView([22.2,79.2],5);
    const mapStatus=document.getElementById("map-status");
    const mapStatus=document.getElementById("map-status");
    const tileLayer=L.tileLayer("https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png",{maxZoom:18,attribution:"&copy; OpenS
| treetMap contributors"}).addTo(map);
    let tileErrors=0;
    const setOfflineMap=(offline)=>{mapStatus.style.display=offline?"block":"none"};
    tileLayer.on("tileload",()=>{tileErrors=0;setOfflineMap(false)});
    tileLayer.on("tileerror",()=>{tileErrors+=1;if(tileErrors>=2)setOfflineMap(true)});
    window.addEventListener("offline",()=>setOfflineMap(true));
    window.addEventListener("online",()=>{tileErrors=0;tileLayer.redraw()});
    if(!navigator.onLine)setOfflineMap(true);
    const state={layers:new Map(),rows:[],sel:null,route:null,routeEnds:[],hist:new Map(),info:new Map(),routeInfo:new Map
| ()};
    const ICONS={jet:"/static/airplane.svg",prop:"/static/propplane.svg",helicopter:"/static/helicopter.svg",default:"/sta
| tic/airplane.svg"};
    const iconFor=(category)=>L.icon({iconUrl:ICONS[category]||ICONS.default,iconSize:[30,30],iconAnchor:[15,15],className
| :"aircraft-icon"});
    function infoFor(icao){const e=state.info.get(icao);return e&&e.status==="done"?e.data:null}
    async function ensureInfo(icao){
      if(state.info.has(icao))return;
      state.info.set(icao,{status:"pending",data:null});
      try{
        const r=await fetch(`/api/aircraft/info?icao=${encodeURIComponent(icao)}`,{cache:"no-store"});
        const p=await r.json().catch(()=>null);
        const data=(p&&p.found)?p:null;
        state.info.set(icao,{status:"done",data});
        const l=state.layers.get(icao);
        if(l&&data&&data.category)l.m.setIcon(iconFor(data.category));
        if(state.sel===icao&&l)panels(l.data);
      }catch(e){
        state.info.set(icao,{status:"error",data:null});
      }
    }
    function routeInfoFor(callsign){const e=state.routeInfo.get(callsign);return e&&e.status==="done"?e.data:null}
    async function ensureRouteInfo(callsign,icao){
      if(state.routeInfo.has(callsign))return;
      state.routeInfo.set(callsign,{status:"pending",data:null});
      try{
        const r=await fetch(`/api/route?callsign=${encodeURIComponent(callsign)}`,{cache:"no-store"});
        const p=await r.json().catch(()=>null);
        const data=(p&&p.found)?p:null;
        state.routeInfo.set(callsign,{status:"done",data});
        if(state.sel===icao){const l=state.layers.get(icao);if(l)panels(l.data)}
      }catch(e){
        state.routeInfo.set(callsign,{status:"error",data:null});
      }
    }
    const els={tracked:document.getElementById("tracked"),visible:document.getElementById("visible"),status:document.getEl
| ementById("status"),list:document.getElementById("aircraftList"),search:document.getElementById("search")};
    const chart=new Chart(document.getElementById("altitudeChart"),{type:"line",data:{labels:[],datasets:[{data:[],borderC
| olor:"#8ecae6",backgroundColor:"rgba(142,202,230,.18)",fill:true,borderWidth:2,pointRadius:1.4,tension:.24}]},options:{m
| aintainAspectRatio:false,plugins:{legend:{display:false}},scales:{x:{ticks:{color:"#9ab7d1"},grid:{color:"rgba(154,183,2
| 09,.15)"}},y:{ticks:{color:"#9ab7d1"},grid:{color:"rgba(154,183,209,.15)"}}}}});
    const num=(v,f=0)=>{const n=Number(v);return Number.isFinite(n)?n:f};
    const M_TO_FT=3.28084,MS_TO_KT=1.943844,MS_TO_FPM=196.8504;
    const norm=(r)=>{const c=String(r.callsign??r.flight_no??"UNKNOWN").trim().toUpperCase()||"UNKNOWN";const i=String(r.i
| caoAddress??r.icao?r.hex??c).trim().toUpperCase()||"UNKNOWN";const lat=num(r.latDD??r.lat??r.latitude,NaN),lon=num(r.lo
| nDD??r.lon??r.longitude,NaN);let a=num(r.altitudeMM?r.altitude??r.alt??0),h=num(r.headingDE2??r.heading??r.track??0),s=
| num(r.horVelocityCMS??r.speed??r.groundspeed??r.gs??0);if(a<100000)a*=1000;if(h<=360)h*=100;if(s<1000)s*=100;return{icao
| Address:i,callsign:c,latDD:lat,lonDD:lon,altitudeMM:a,headingDE2:h,horVelocityCMS:s,verVelocityCMS:num(r.verVelocityCMS?
| ?0,0),squawk:num(r.squawk??0,0),sourceIp:String(r.sourceIp??"").trim(),isPredicted:Boolean(r.isPredicted),predictionAgeS
| ec:num(r.predictionAgeSec??0,0)};};
    const pop=(a)=>`<b>${a.callsign}</b>${a.isPredicted?' <span style="color:#ff7d7d">(Predicted)</span>':''}<br>ICAO: ${a
| .icaoAddress}<br>ALT: ${Math.round(a.altitudeMM/1000)} m<br>SPD: ${Math.round(a.horVelocityCMS/100)} m/s<br>HDG: ${Math.
| round(a.headingDE2/100)} deg${a.isPredicted?`<br>PRED AGE: ${Math.round(a.predictionAgeSec)} s`:''}`;
    const haversine=(a,b)=>{const R=6371000,rad=v=>v*Math.PI/180,p1=rad(a[0]),p2=rad(b[0]),dLat=rad(b[0]-a[0]),dLon=rad(b[
| 1]-a[1]);const h=Math.sin(dLat/2)**2+Math.cos(p1)*Math.cos(p2)*Math.sin(dLon/2)**2;return 2*R*Math.atan2(Math.sqrt(h),Ma
| th.sqrt(1-h));};
    const PLACES=[
      {name:"Delhi",lat:28.6139,lon:77.2090},{name:"Mumbai",lat:19.0760,lon:72.8777},{name:"Bengaluru",lat:12.9716,lon:77.
| 5946},
    function clearRoute(){if(state.route){map.removeLayer(state.route);state.route=null}state.routeEnds.forEach(m=>map.rem
| oveLayer(m));state.routeEnds=[]}
    function drawRoute(a){clearRoute();const l=state.layers.get(a.icaoAddress);if(!l)return;const live=l.tr.getLatLngs()||
| [];const pred=l.pr.getLatLngs()||[];const all=[...live,...pred].map(p=>[p.lat,p.lng]);if(all.length<2)return;state.route
| =L.polyline(all,{color:"#fdb813",weight:2,dashArray:"5 6",opacity:.9}).addTo(map);const f=L.circleMarker(all[0],{radius:
| 4,color:"#fdb813",fillColor:"#fdb813",fillOpacity:1}).addTo(map);const t=L.circleMarker(all[all.length-1],{radius:4,colo
| r:"#ff7d7d",fillColor:"#ff7d7d",fillOpacity:1}).addTo(map);state.routeEnds=[f,t]}
    function flightStatus(vsFpm){if(vsFpm>150)return{label:"Climbing",cls:"fbadge-climb",arrow:"?"};if(vsFpm<-150)return{l
| abel:"Descending",cls:"fbadge-descend",arrow:"?"};return{label:"Level",cls:"fbadge-level",arrow:"?"};}
    function panels(a){
      const altM=Math.round(a.altitudeMM/1000),altFt=Math.round(altM*M_TO_FT);
      const spdMs=Math.round(a.horVelocityCMS/100),spdKt=Math.round(spdMs*MS_TO_KT);
      const vsMs=a.verVelocityCMS/100,vsFpm=Math.round(vsMs*MS_TO_FPM);
      const hdg=Math.round(a.headingDE2/100);
```